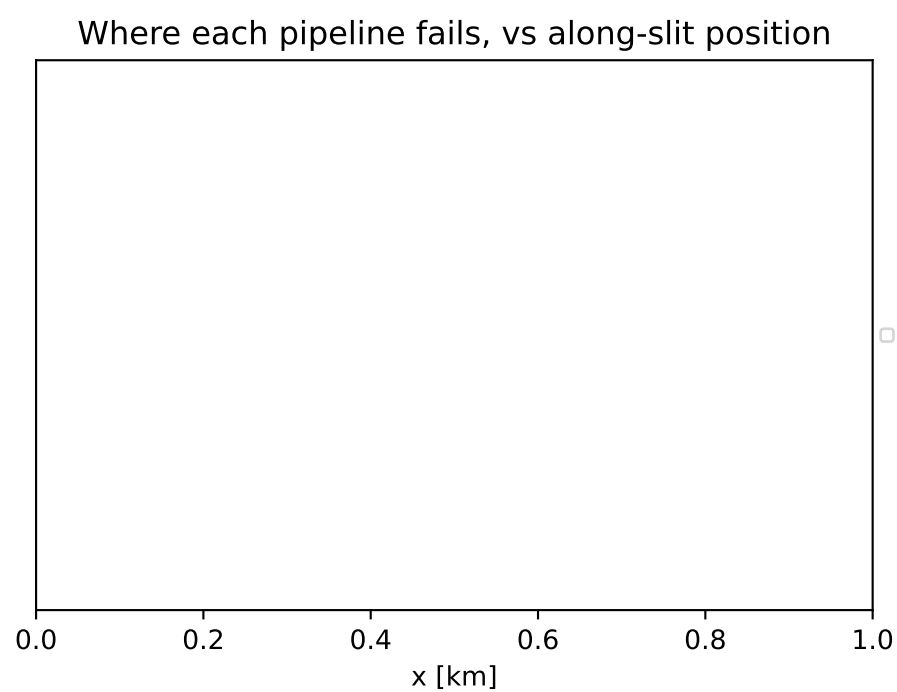
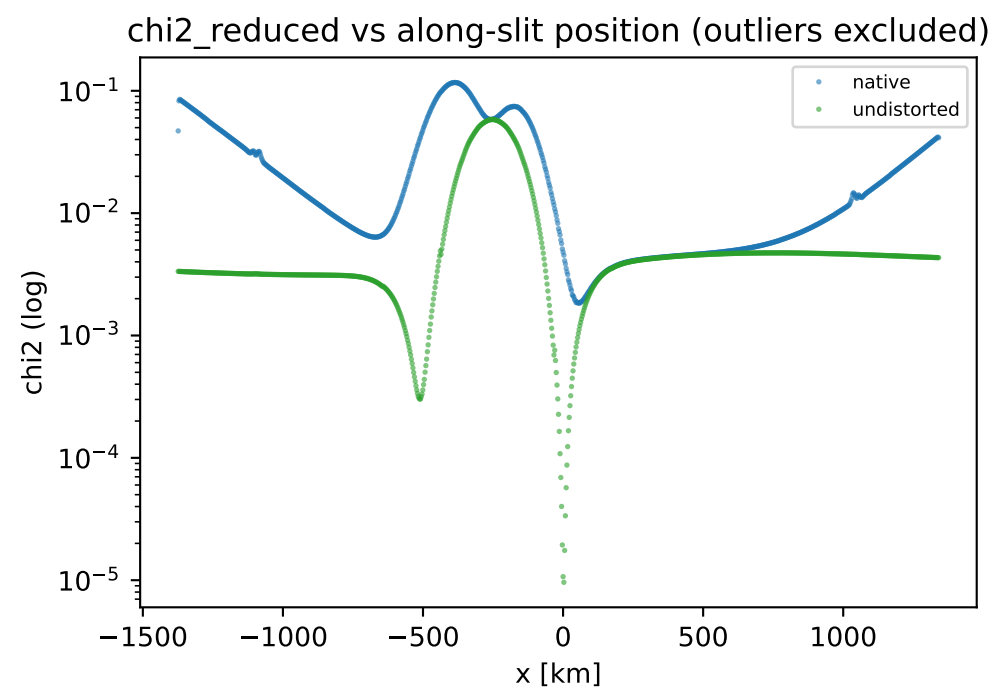
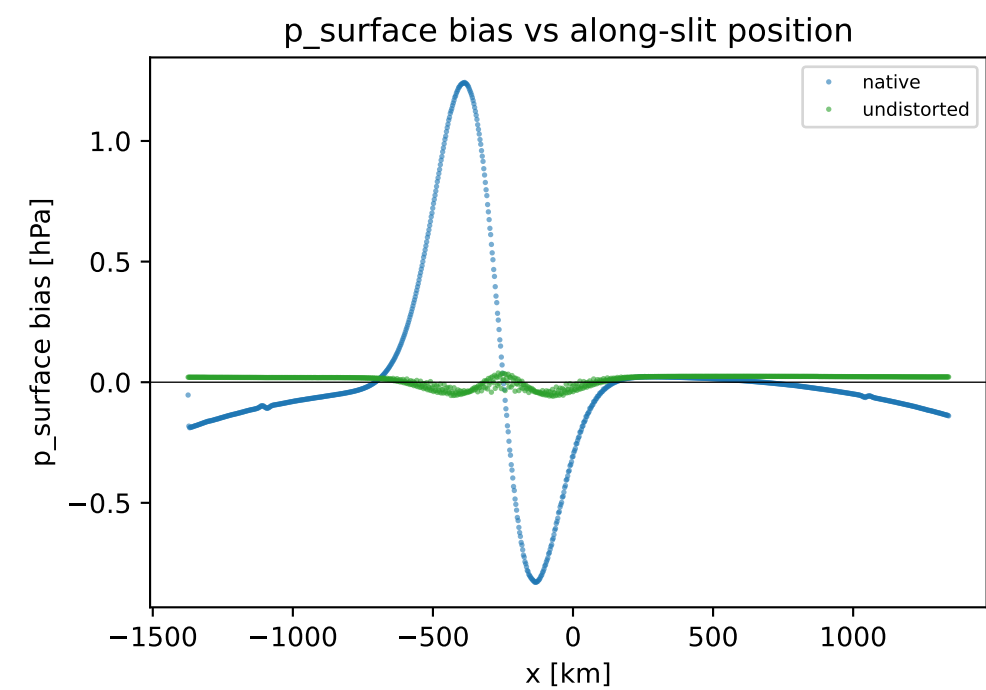
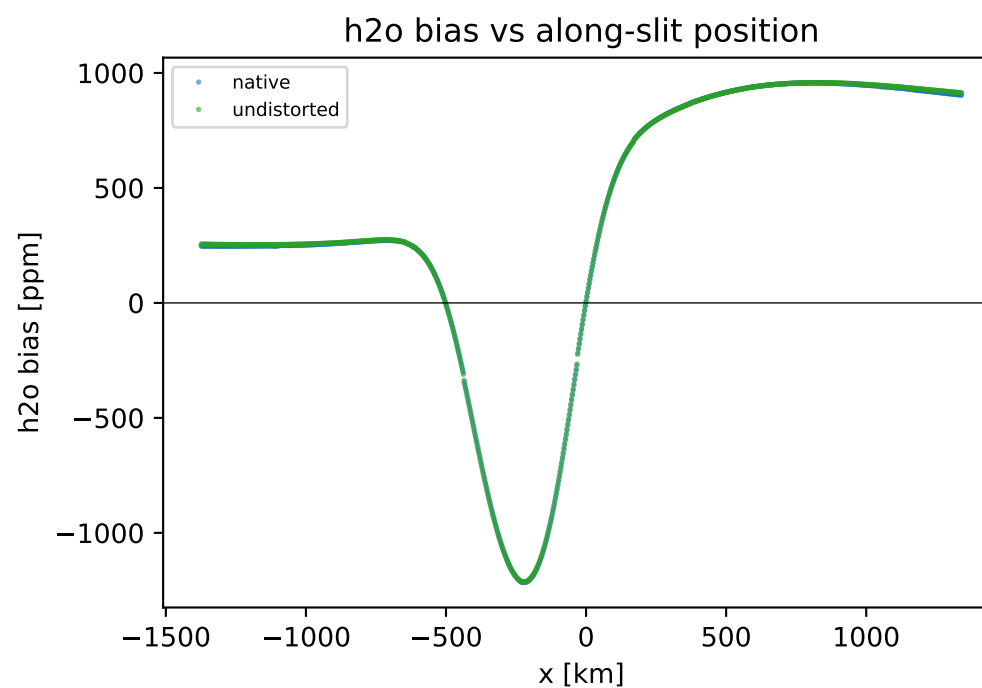
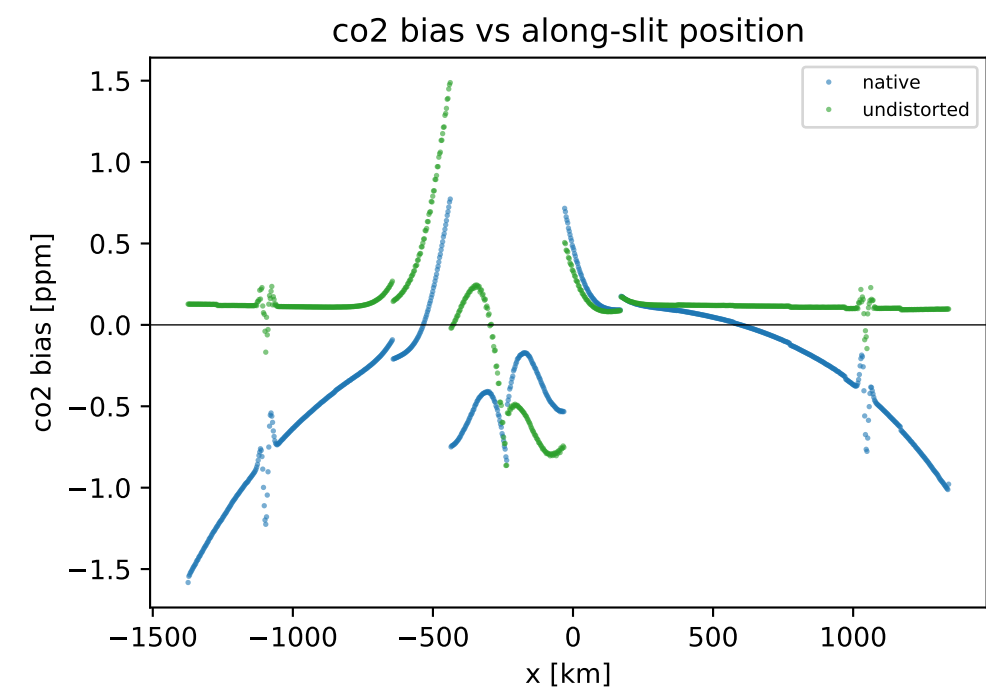
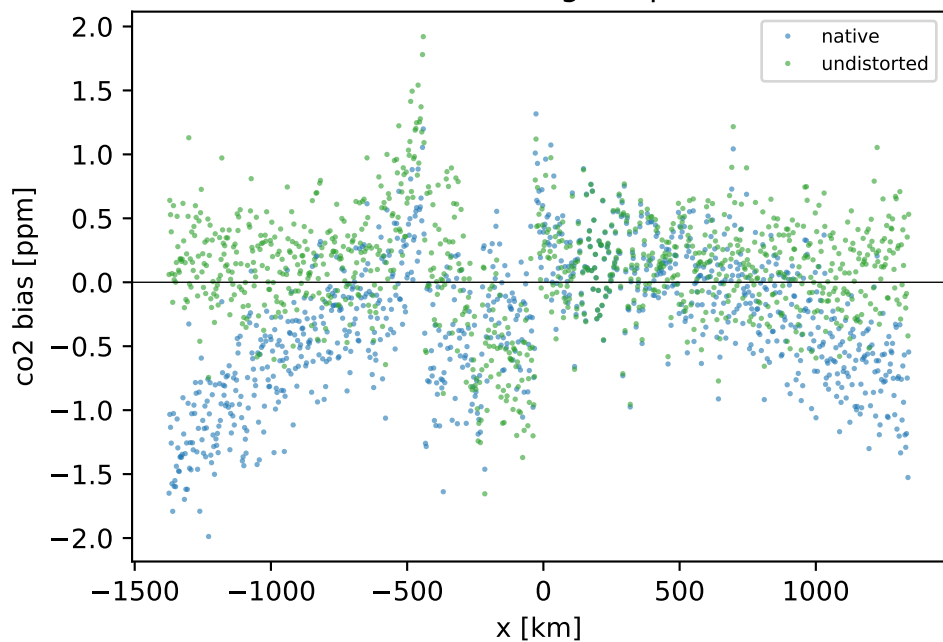


Joint FPA0+FPA1 -- realistic, no noise (native/undistorted)

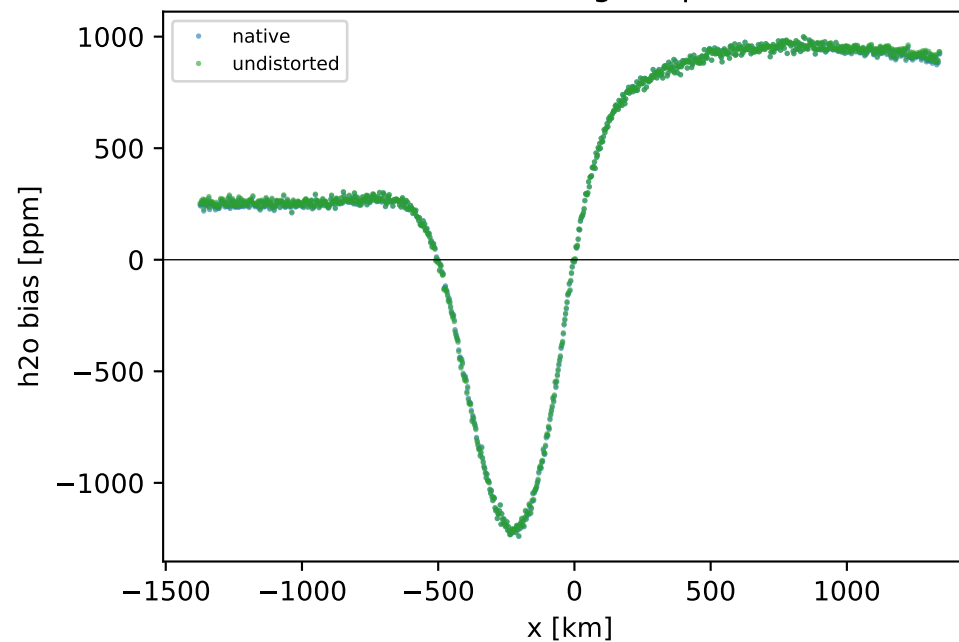


Joint FPA0+FPA1 -- realistic, noise (native/undistorted)

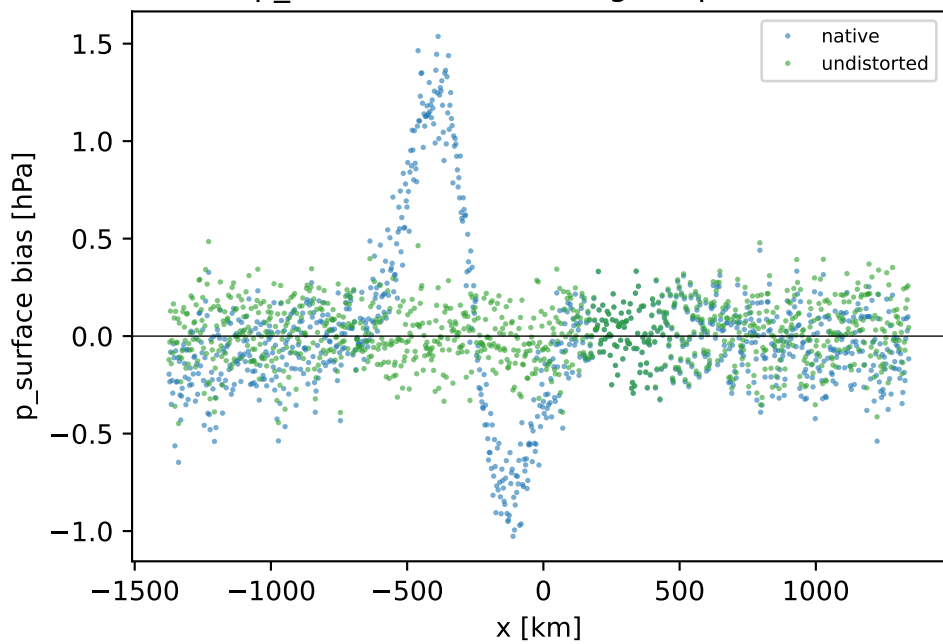
co2 bias vs along-slit position



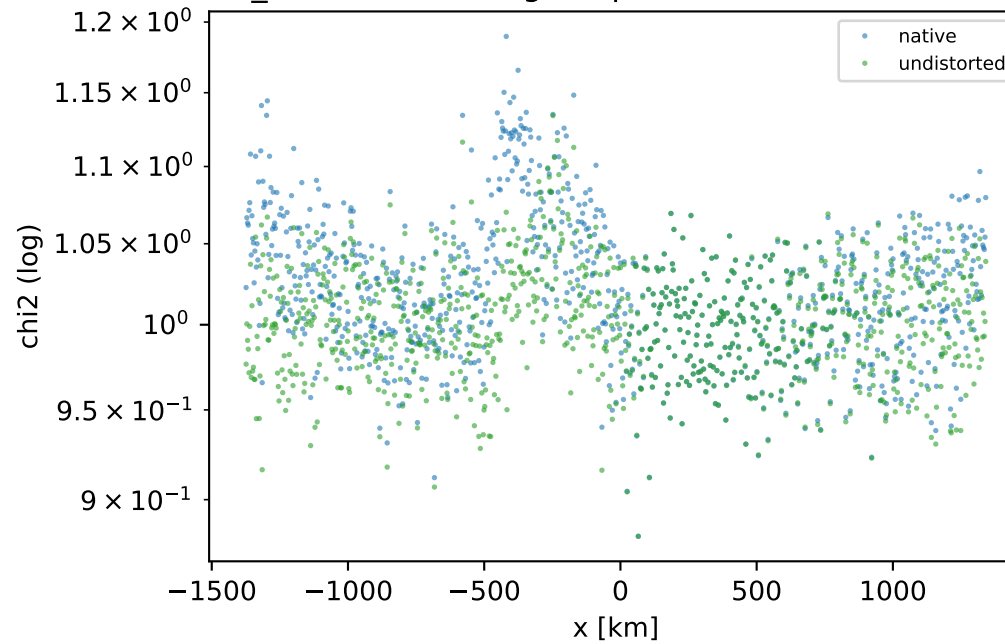
h2o bias vs along-slit position



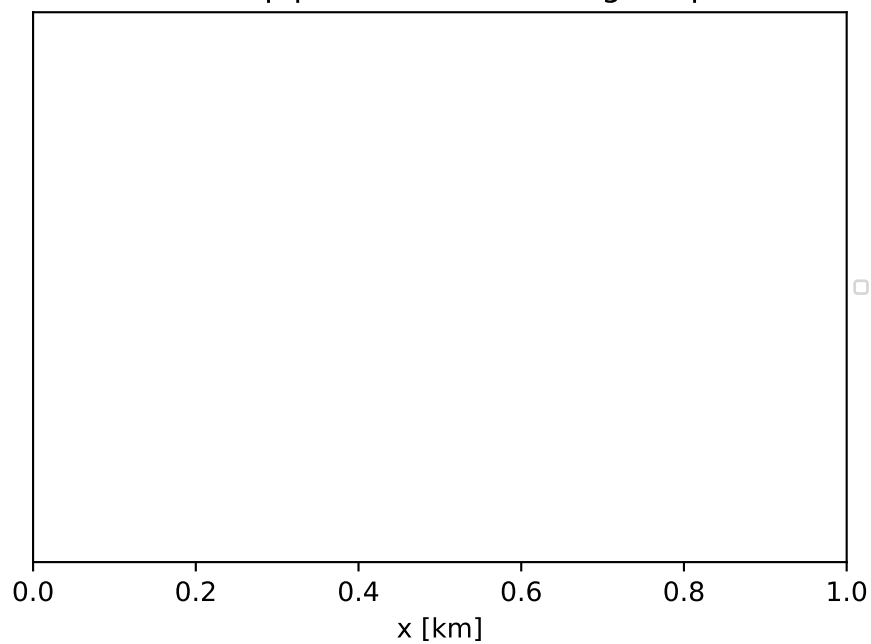
p_surface bias vs along-slit position



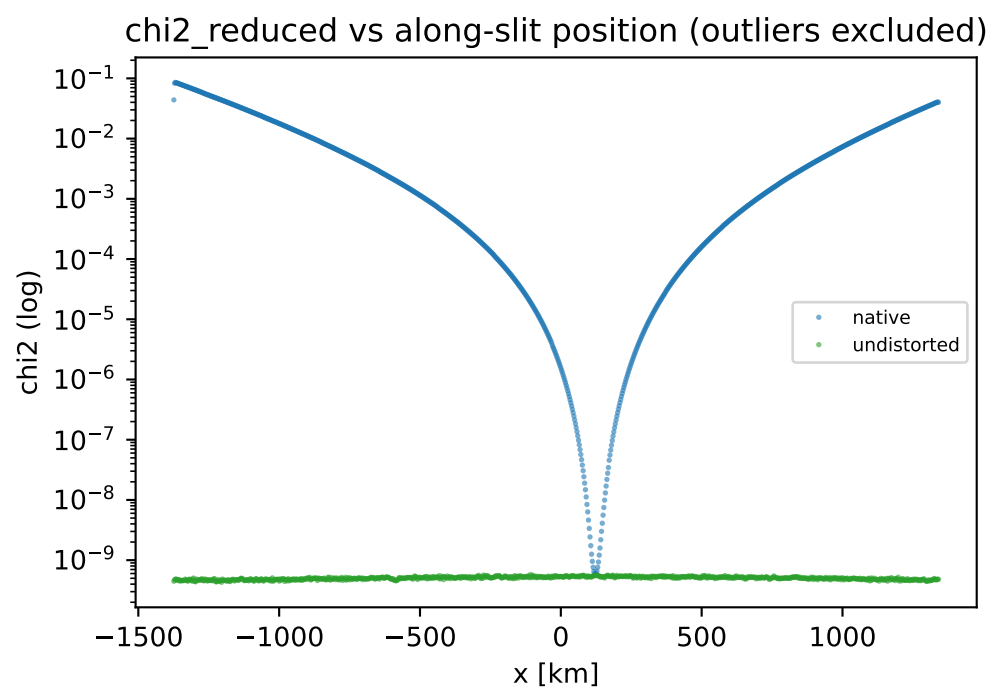
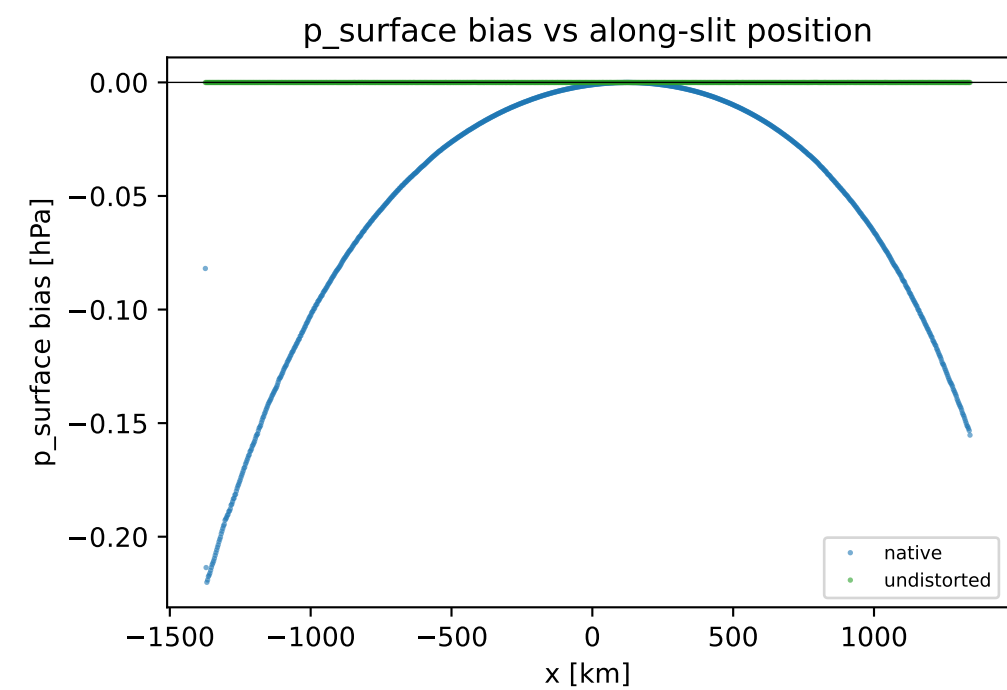
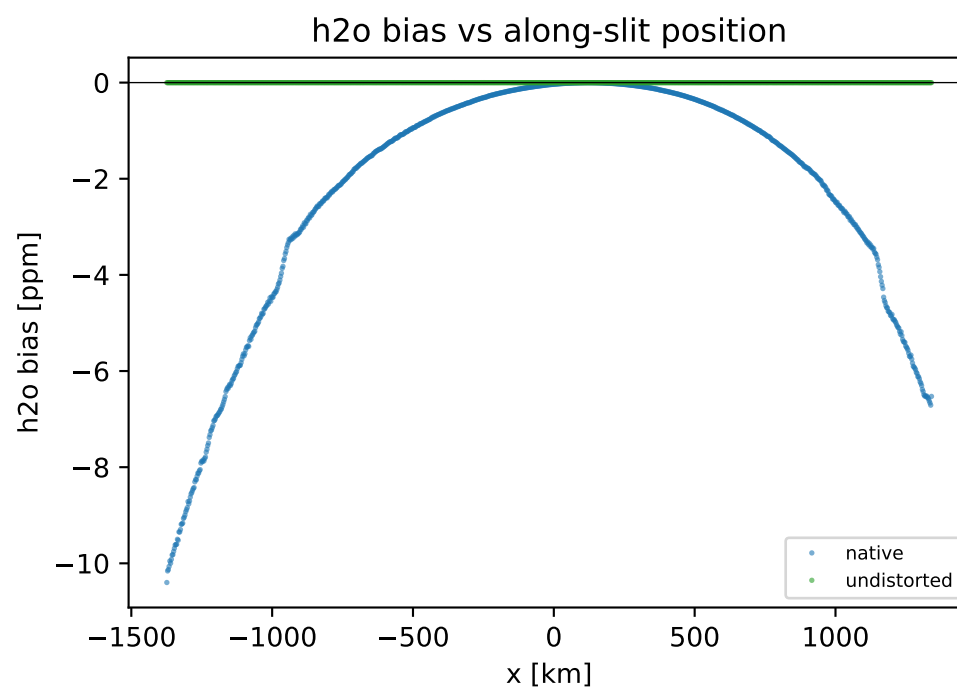
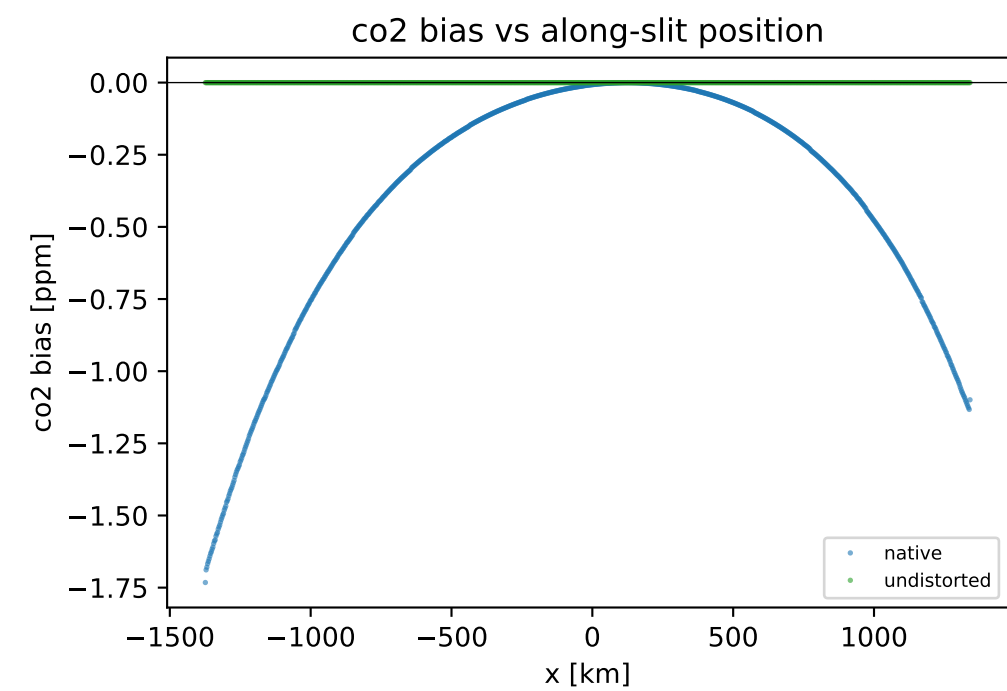
chi2_reduced vs along-slit position (outliers excluded)



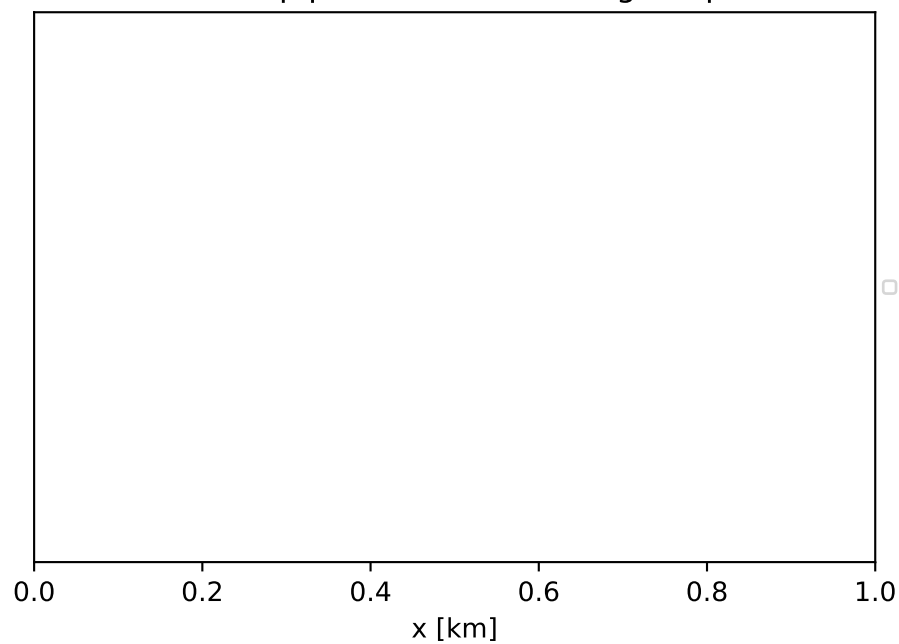
Where each pipeline fails, vs along-slit position



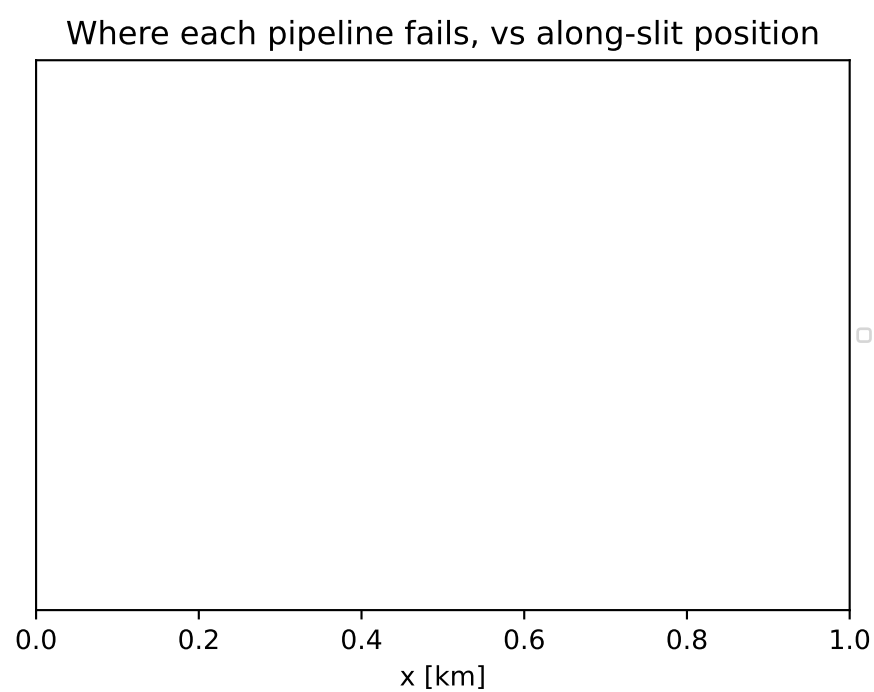
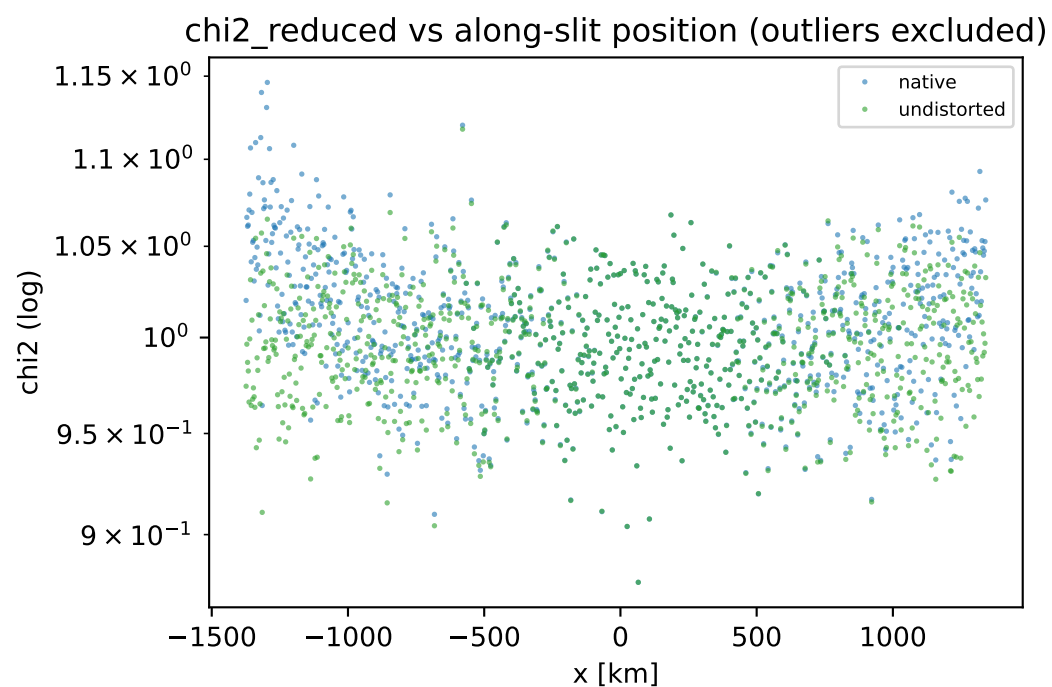
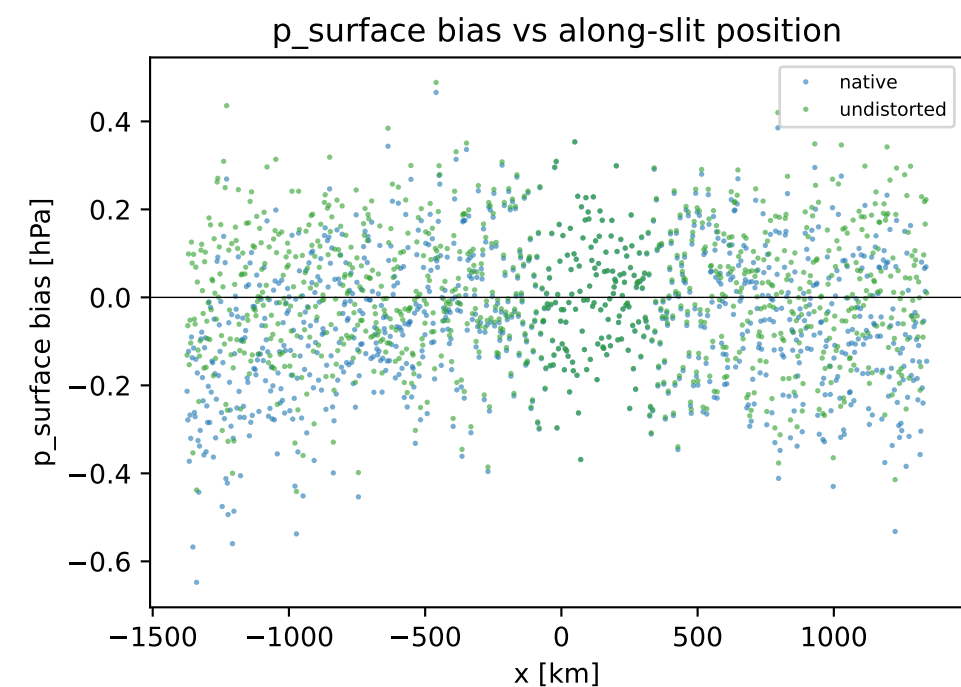
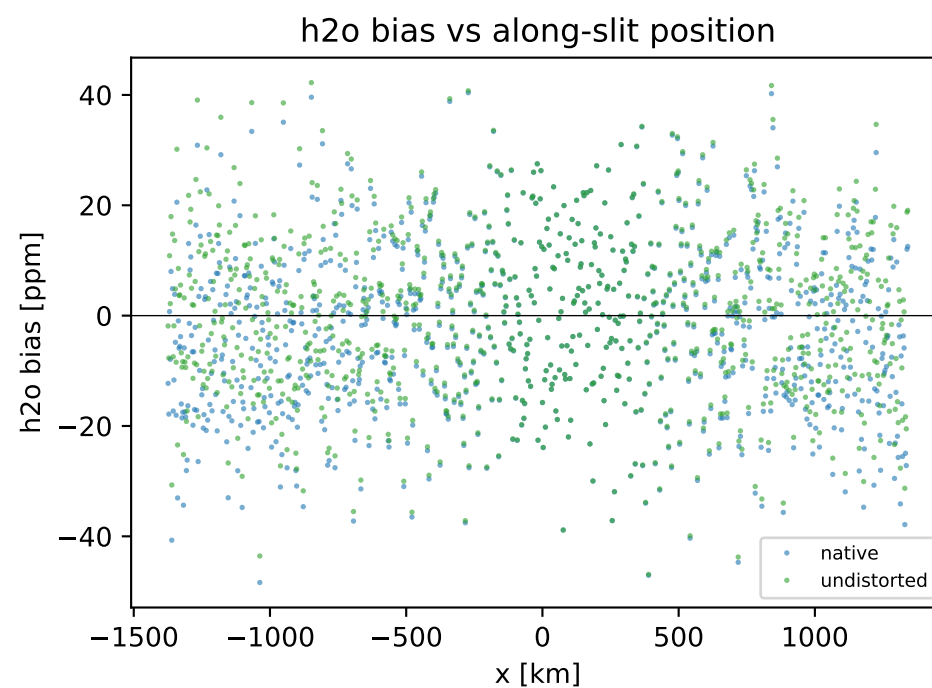
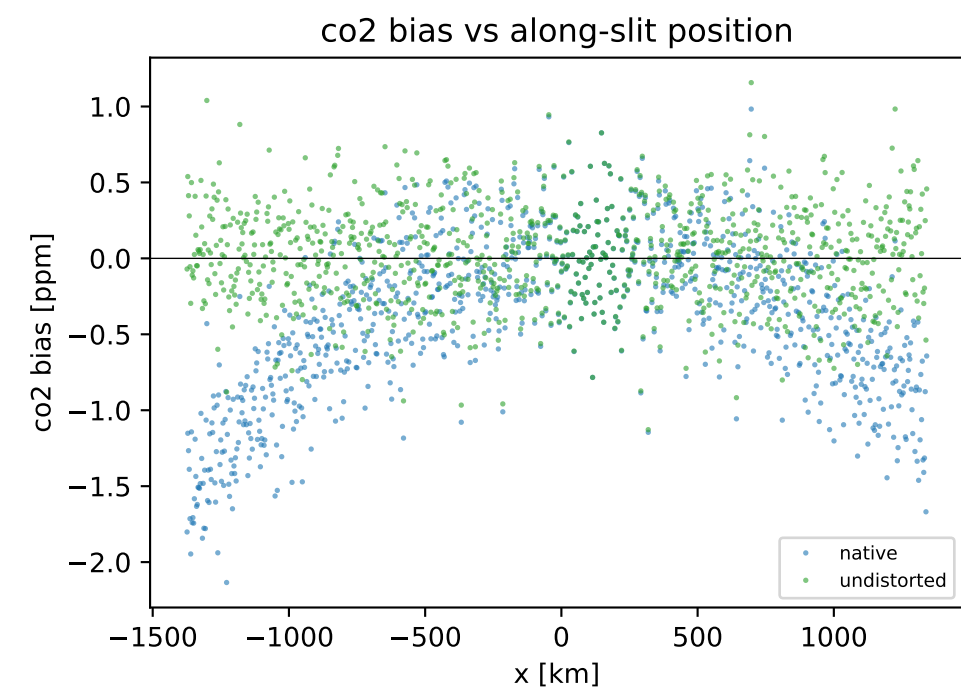
Joint FPA0+FPA1 -- uniform, no noise (native/undistorted)



Where each pipeline fails, vs along-slit position

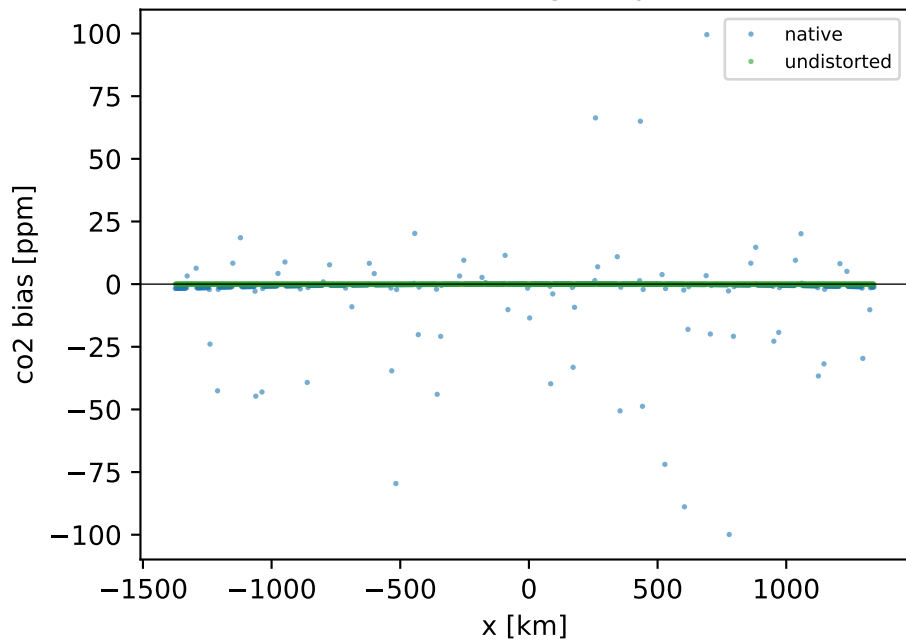


Joint FPA0+FPA1 -- uniform, noise (native/undistorted)

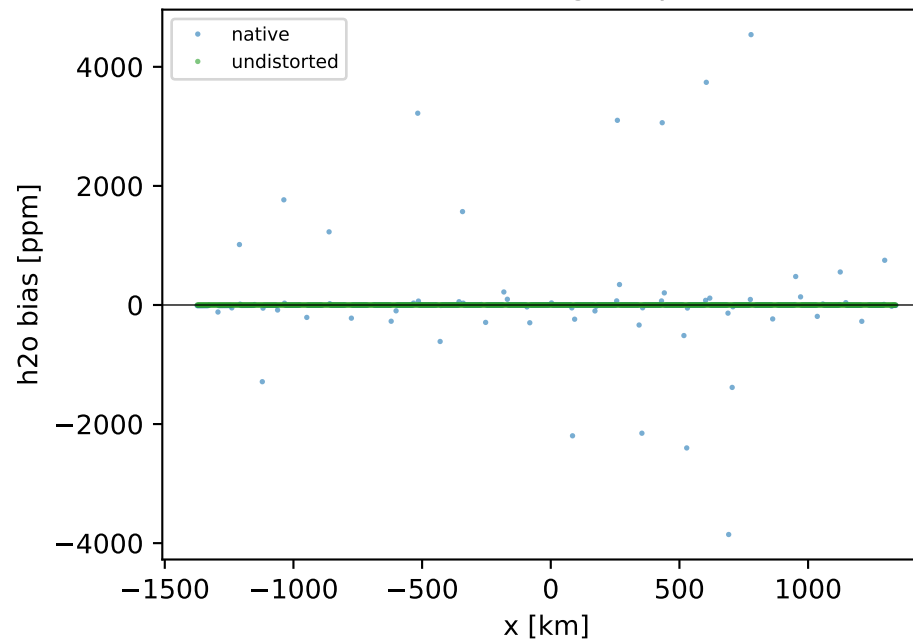


Joint FPA0+FPA1 -- barcode, no noise (native/undistorted)

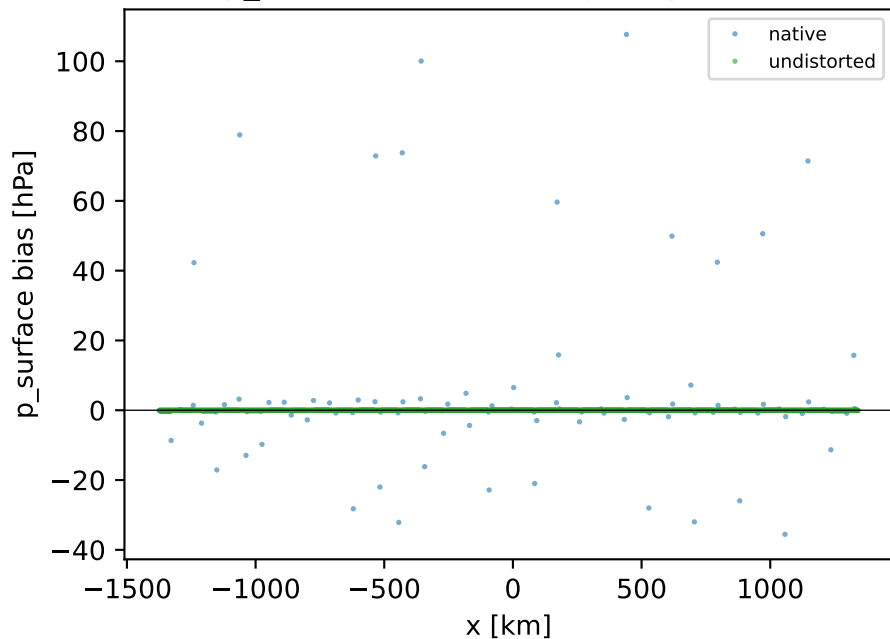
co2 bias vs along-slit position



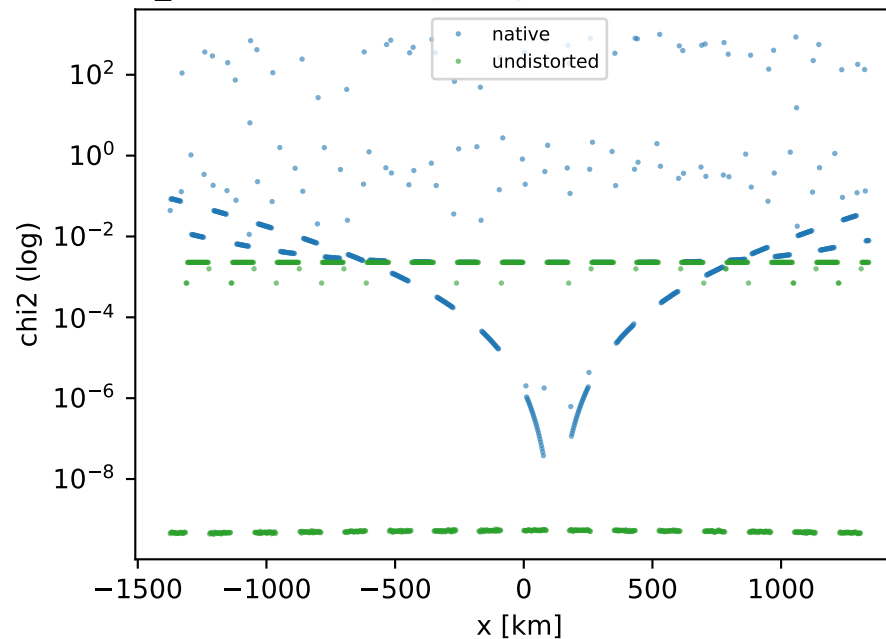
h2o bias vs along-slit position



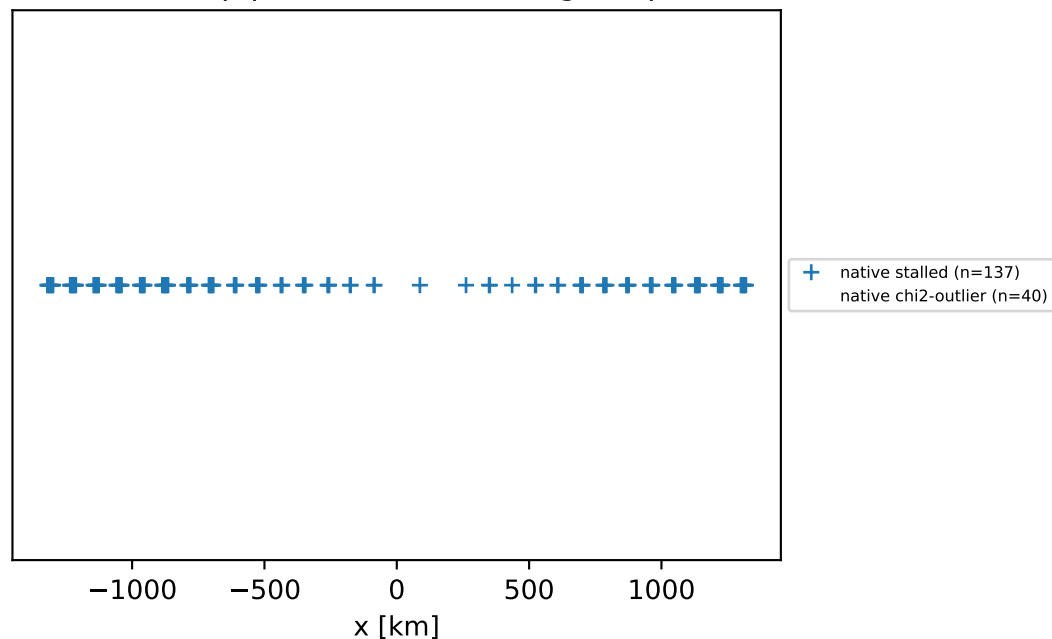
p_surface bias vs along-slit position



chi2_reduced vs along-slit position (outliers excluded)

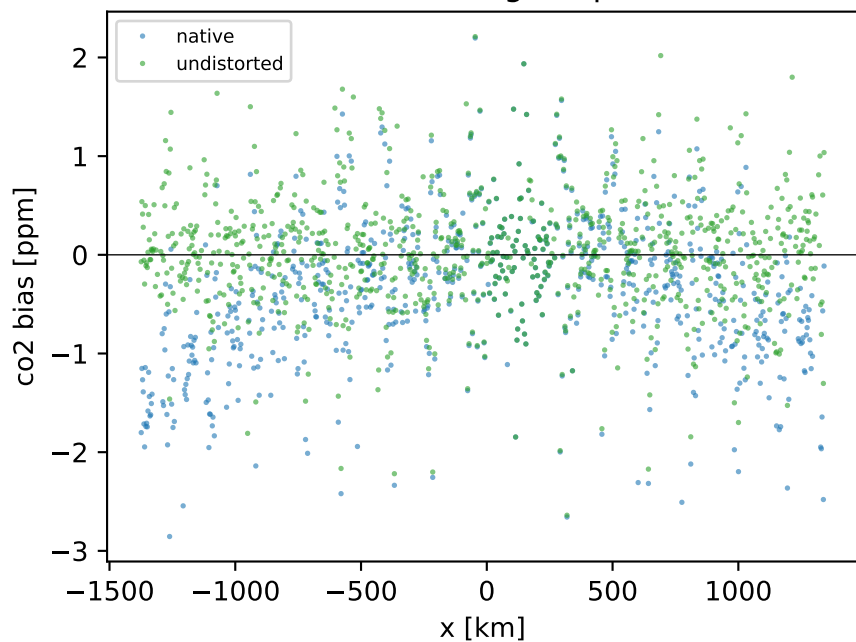


Where each pipeline fails, vs along-slit position

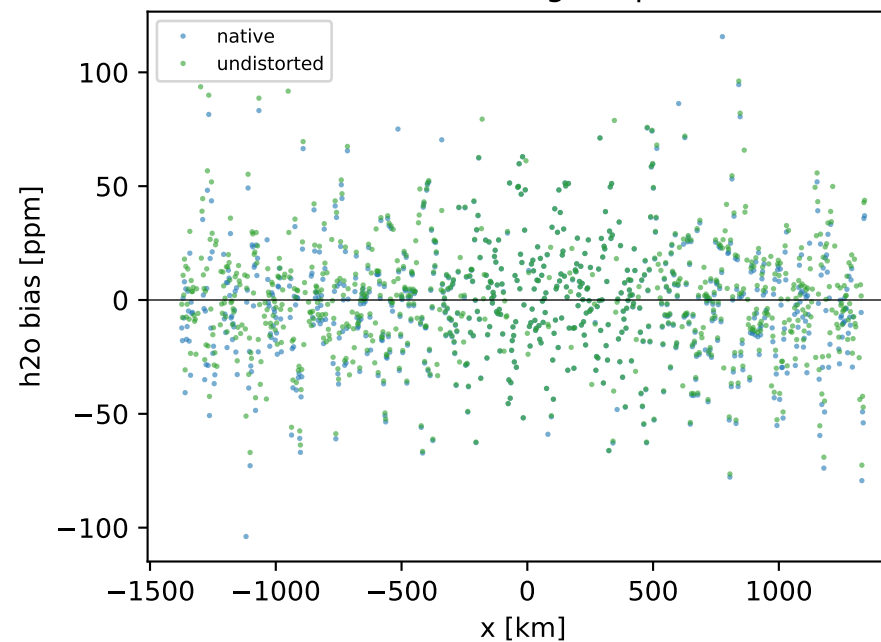


Joint FPA0+FPA1 -- barcode, noise (native/undistorted)

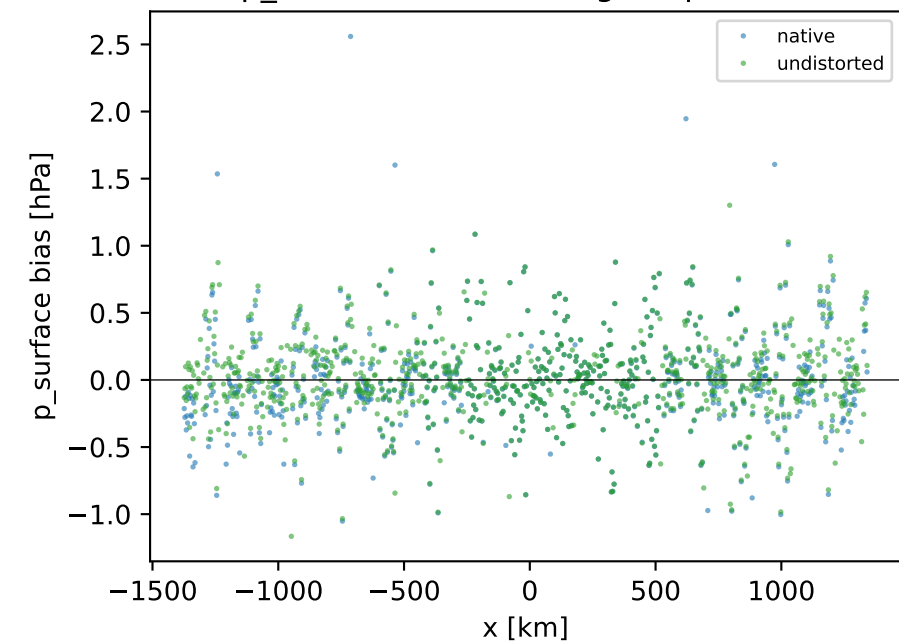
co2 bias vs along-slit position



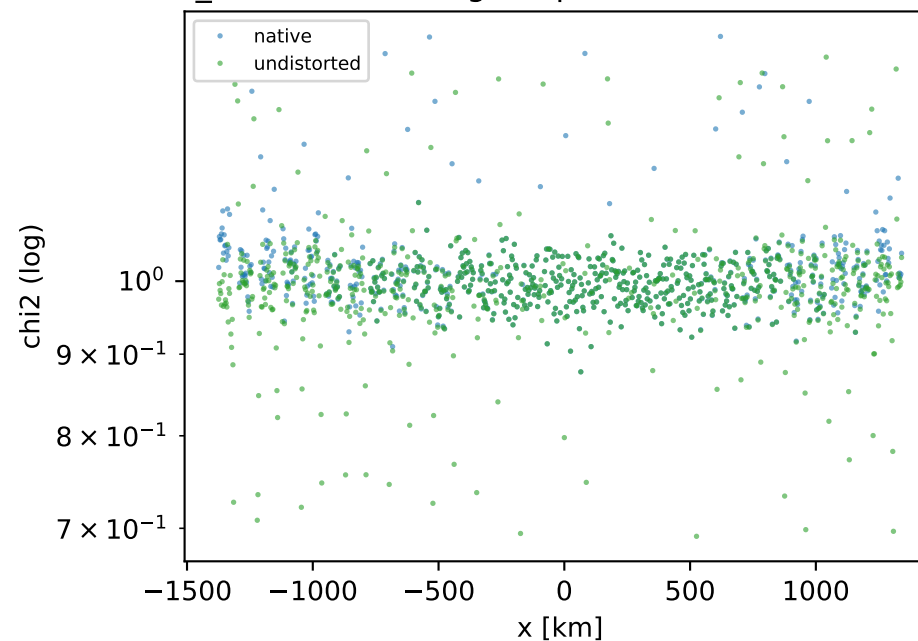
h2o bias vs along-slit position



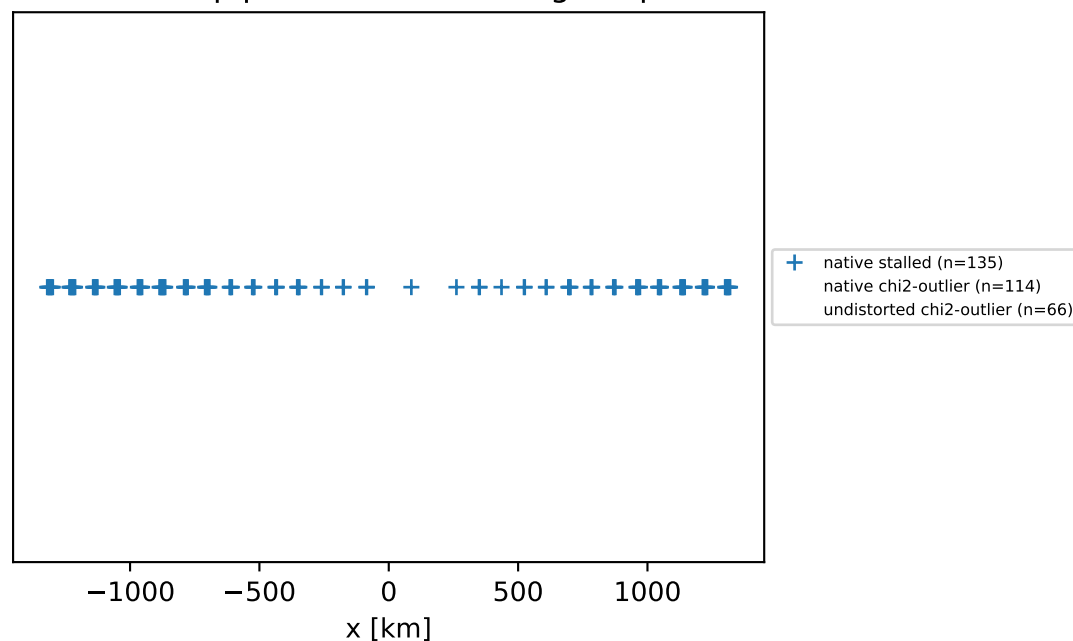
p_surface bias vs along-slit position



chi2_reduced vs along-slit position (outliers excluded)



Where each pipeline fails, vs along-slit position



FPA0+FPA1 -- robust bias statistics across scenes/noise (native/undistorted)

scene	noise	quantity	pipeline	n	mean	std	median	MAD-sig	n_out
realistic	0	co2	native	1002	-0.331	0.4377	-0.2727	0.4568	0
realistic	0	h2o	native	1002	+362.7	623.9	+270.9	891.6	0
realistic	0	p_surface	native	1002	+0.009817	0.3654	-0.03659	0.08442	112
realistic	0	co2	undistorted	1002	+0.09014	0.2931	+0.1154	0.01865	175
realistic	0	h2o	undistorted	1002	+364.8	624.3	+273.3	886.6	0
realistic	0	p_surface	undistorted	1002	+0.01021	0.02155	+0.01982	0.00586	113
realistic	1	co2	native	1002	-0.3104	0.538	-0.2801	0.5589	0
realistic	1	h2o	native	1002	+362.8	623.8	+280.6	875.9	0
realistic	1	p_surface	native	1002	+0.01359	0.4004	-0.02749	0.2254	0
realistic	1	co2	undistorted	1002	+0.1107	0.4344	+0.134	0.3545	0
realistic	1	h2o	undistorted	1002	+364.9	624.3	+283.2	870.7	0
realistic	1	p_surface	undistorted	1002	+0.014	0.1528	+0.01155	0.1579	0
uniform	0	co2	native	1002	-0.4003	0.4199	-0.2435	0.3288	0
uniform	0	h2o	native	1002	-2.241	2.519	-1.227	1.669	0
uniform	0	p_surface	native	1002	-0.05424	0.05565	-0.03386	0.04552	0
uniform	0	co2	undistorted	1002	-0.000164	3.818e-06	-0.0001638	3.746e-06	0
uniform	0	h2o	undistorted	1002	-0.001045	0.0001167	-0.001042	0.0001256	0
uniform	0	p_surface	undistorted	1002	-3.669e-05	6.85e-06	-3.871e-05	5.436e-06	0
uniform	1	co2	native	1002	-0.3792	0.5219	-0.3087	0.5073	0
uniform	1	h2o	native	1002	-2.116	15.02	-1.637	15.68	0
uniform	1	p_surface	native	1002	-0.05063	0.1592	-0.04781	0.1628	0
uniform	1	co2	undistorted	1002	+0.02094	0.3208	+0.02474	0.3195	0
uniform	1	h2o	undistorted	1002	+0.1239	14.76	+0.7134	15.89	0
uniform	1	p_surface	undistorted	1002	+0.003608	0.1486	+0.0003234	0.1504	0
barcode	0	co2	native	825	-1.151	9.521	-0.2155	0.3093	60
barcode	0	h2o	native	825	+8.974	362.6	-0.9918	1.439	64
barcode	0	p_surface	native	825	+0.5487	8.844	-0.01379	0.01982	105
barcode	0	co2	undistorted	1002	-0.0001494	1.651e-05	-0.0001558	2.263e-05	0
barcode	0	h2o	undistorted	1002	-0.0001465	0.0009291	-0.0008188	0.0007285	0
barcode	0	p_surface	undistorted	1002	-1.669e-05	2.176e-05	-1.592e-05	3.16e-05	0
barcode	1	co2	native	753	-0.3657	0.728	-0.3004	0.6483	0
barcode	1	h2o	native	753	-1.313	27	-0.9526	21.51	0
barcode	1	p_surface	native	753	-0.002072	0.3553	-0.03566	0.241	2
barcode	1	co2	undistorted	936	+0.02906	0.5855	+0.02628	0.4426	0
barcode	1	h2o	undistorted	936	+0.8823	25.39	+0.7343	20.65	0
barcode	1	p_surface	undistorted	936	+0.02028	0.3094	+0.003896	0.2236	0